CAPSTONE PROJECT DOCUMENTATION

---

1. PROJECT OVERVIEW

Project Title:
FoodMart Delivery Performance & Customer Behavior Analysis

Domain:
Food Delivery / E-Commerce / Data Analytics

Problem Statement:
Food delivery platforms generate large volumes of data related to orders, customers, restaurants, delivery partners, and complaints. However, without proper analysis, it becomes difficult to identify performance issues such as delivery delays, low customer ratings, and restaurant inefficiencies. There is a need for a data-driven solution to analyze delivery performance, customer behavior, and operational challenges to improve service quality and business efficiency.

Objective of the Project:

- Analyze delivery performance and customer behavior
- Identify peak order periods and revenue trends
- Evaluate restaurant and delivery partner efficiency
- Detect issues affecting customer satisfaction
- Provide actionable insights using interactive dashboards

Business Use Case:
This project helps food delivery companies optimize delivery operations, improve customer satisfaction, reduce delays, and enhance restaurant performance through data-driven decision-making.

2. DATASET DESCRIPTION (EXCEL)

Data Source:

Internal food delivery platform dataset (CSV files imported from MySQL)

Number of Records:

- Customers: 10,000
- Restaurants: 500
- Delivery Partners: 1,000
- Orders: 100,000
- Complaints: ~5,000

Number of Columns:

Approximately 25+ columns across all tables

Column Names:

- Orders: order_id, customer_id, restaurant_id, partner_id, order_date, order_status, order_value, delivery_time, prep_time, rating
- Customers: customer_id, name, city, signup_date
- Restaurants: restaurant_id, restaurant_name, cuisine_type, city
- Delivery Partners: partner_id, partner_name, joining_date
- Complaints: complaint_id, order_id, complaint_type, resolution_time

Description of Dataset:

The dataset represents end-to-end food delivery operations, capturing order transactions, customer details, restaurant information, delivery partner performance, and customer complaints.

Data Cleaning Steps Performed:

- Removed duplicate records
- Handled missing values
- Standardized date and numeric formats
- Sorted and filtered data for analysis

Final Outcome of Excel Work:

- Cleaned and structured datasets
- Pivot tables for order trends, delivery performance, and complaints
- Exploratory charts for preliminary insights

3. SQL IMPLEMENTATION

Database Name:
Capstone

Table Name(s):
Customers, Restaurants, Delivery_Partners, Orders, Complaints

Purpose of Using SQL:

- Store and manage large datasets efficiently
- Perform data extraction and aggregation
- Answer business-related analytical questions

Types of Queries Used:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- Subqueries

Business Questions Solved Using SQL:

1. Which cities have the highest number of orders and delays?
2. Which restaurants generate the highest revenue?
3. Which delivery partners have the best and worst performance?

Key Insights from SQL Queries:

- Certain cities contribute significantly to total orders and delays
- A small number of restaurants generate a large share of revenue
- Delivery partner performance varies widely based on delivery time

4. POWER BI REPORT & VISUALIZATION

Dashboard Objective:

To provide a one-page interactive dashboard that gives insights into order trends, revenue, delivery performance, customer satisfaction, and operational inefficiencies.

KPIs Used:

- Total Orders
- Total Revenue
- Average Delivery Time
- Average Rating

Charts Used:

- Bar Chart (Top cuisines, restaurants, delivery partners)
- Line Chart (Monthly revenue trend)
- Pie / Donut Chart (Cuisine and complaint distribution)
- Cards (KPI tiles)

Filters / Slicers Used:

- City
- Cuisine Type
- Month

Key Insights from Power BI Dashboard:

- Peak order months were clearly identified
- Slow-performing delivery partners were flagged
- Restaurants with high cancellation rates require review
- Longer delivery times negatively impact customer ratings

Business Decisions Supported by Dashboard:

- Optimize delivery partner allocation
- Improve restaurant service quality
- Reduce delivery delays during peak months
- Enhance customer satisfaction strategies

5. OVERALL INSIGHTS & FINDINGS

Major Insights Identified:

- Delivery delays strongly affect customer ratings
- Few restaurants dominate revenue generation
- Certain cities experience consistent delivery issues

Patterns / Trends Observed:

- Seasonal spikes in monthly revenue
- Higher complaints during peak order periods
- Faster delivery partners receive better ratings

Recommendations Based on Analysis:

- Monitor low-performing delivery partners
- Improve logistics in high-delay cities
- Focus on quality control for high-volume restaurants

6. CONCLUSION

Summary of Project Work:
This project successfully extracted, cleaned, analyzed, and visualized a large food delivery dataset using SQL, Excel, and Power BI. It delivered actionable insights into delivery efficiency, customer satisfaction, and business performance.

Tools Used:

- Excel
- MySQL
- Power BI

Skills Gained:

- Data cleaning and preprocessing
- SQL querying and database management
- Data visualization and dashboard design
- Business insight generation

Real-Time Industry Relevance:
The project reflects real-world food delivery analytics use cases and supports operational and strategic decision-making in e-commerce platforms.

7. DECLARATION

I hereby declare that this project is my original work and has not been copied from any source.

Student Name: Anish V S
Register Number: 23106006
Date: 31-12-2025